CHAPTER 2 · POLYNOMIALS

10 Diverse Mock Test Sets · 100 Questions with Answers and Explanations

The original question order and question wording have been retained. Each answer and explanation is placed immediately beneath its question. MCQ answers identify the correct option.

MOCK SET 01 | Core concepts

Q1. [Quick MCQ · Easy] If 2 is a zero of $p(x)=x^2-5x+k$, find k. (A) 2 (B) 4 (C) 6 (D) 10 (1 mark)

Correct answer: (C) 6

Explanation: Since 2 is a zero, $p(2)=0$. Thus $2^2-5(2)+k=0$, so $4-10+k=0$. Therefore $k=6$.

Q2. [Spot the error · Easy] A student says the degree of $5x^2+3x\blacksquare-7$ is 2 because $5x^2$ is written first. Is the student correct? (1 mark)

Answer: No.

Explanation: The degree is the greatest exponent of x with a non-zero coefficient. Here the greatest exponent is 7, so the degree is 7.

Q3. [Skill check · Moderate] Find the zeros of $x^2-9x+20$. (1 mark)

Answer: 4 and 5.

Explanation: Factorise: $x^2-9x+20=(x-4)(x-5)$. Setting each factor equal to zero gives $x=4$ or $x=5$.

Q4. [One-line · Easy] How many real zeros can a quadratic polynomial have at most? (1 mark)

Answer: 2 real zeros.

Explanation: A non-zero polynomial of degree 2 can have at most 2 zeros.

Q5. [Reasoning · Moderate] A quadratic has zeros −2 and 7. Find its monic polynomial. (1 mark)

Answer: $x^2-5x-14$.

Explanation: Sum of zeros = $-2+7=5$ and product = $(-2)(7)=-14$. The monic quadratic is $x^2-(\text{sum})x+(\text{product})=x^2-5x-14$.

Q6. [Word problem · Moderate] A rectangular display has area $x^2+7x+12$. Its side expressions are $x+3$ and $x+4$. What are the zeros of its area polynomial? (1 mark)

Answer: −3 and −4.

Explanation: $x^2+7x+12=(x+3)(x+4)$. Setting each factor to zero gives $x=-3$ and $x=-4$.

Q7. [PYQ-theme · Moderate] For $p(x)=3x^2-8x+4$, find the sum and product of its zeros. (1 mark)

Answer: Sum = 8/3; product = 4/3.

Explanation: For ax^2+bx+c, sum of zeros = $-b/a$ and product = c/a. Here $a=3$, $b=-8$, $c=4$. Hence sum=8/3 and product=4/3.

Q8. [Challenge · Hard] If α and β are zeros of $2x^2-7x+3$, find $\alpha^2+\beta^2$ without solving for α and β. (3 marks)

Answer: 37/4.

Explanation: Sum $\alpha+\beta=7/2$ and product $\alpha\beta=3/2$. Use $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$. Therefore $\alpha^2+\beta^2=(7/2)^2-2(3/2)=49/4-12/4=37/4$.

Q9. [Subjective · 3 marks] Find the zeros of $2x^2-x-6$ and verify the relationships between zeros and coefficients. (3 marks)

Answer: 2 and −3/2.

Explanation: Factorise: $2x^2-x-6=(2x+3)(x-2)$. Thus the zeros are 2 and −3/2. Their sum is $2-3/2=1/2=-b/a=-(-1)/2$. Their product is $2(-3/2)=-3=c/a=-6/2$. Both relationships are verified.

Q10. [Subjective proof · 3 marks] If one zero of $x^2+(k-3)x-10$ is 2, find k and the other zero. Show your steps. (4 marks)

Answer: k=6; the other zero is −5.

Explanation: Substitute x=2: $2^2+(k-3)(2)-10=0$. So $4+2k-6-10=0$, giving $2k-12=0$ and k=6. The product of the zeros is −10. If one zero is 2, the other is $-10/2=-5$.

MOCK SET 02 | Graphs & translation

Q1. [Graph reading · Easy] A polynomial graph crosses the x-axis at x=−1 and x=4. State its zeros. (1 mark)

Answer: −1 and 4.

Explanation: The zeros are the x-coordinates where the graph meets or crosses the x-axis.

Q2. [MCQ · Moderate] Which quadratic has zeros 3 and −2? (A) x^2-x-6 (B) x^2+x-6 (C) x^2-5x+6 (D) x^2+5x+6 (1 mark)

Correct answer: (A) x^2-x-6

Explanation: A monic quadratic with zeros 3 and −2 is $(x-3)(x+2)=x^2-x-6$. Therefore option (A) is correct.

Q3. [Find the parameter · Moderate] If the zeros of $x^2-(m+1)x+12$ are 3 and 4, find m. (1 mark)

Answer: m=6.

Explanation: The sum of the zeros is 3+4=7. For $x^2-(m+1)x+12$, the sum is m+1. Thus m+1=7, so m=6.

Q4. [True/False with reason · Moderate] If the sum and product of zeros of a monic quadratic are 0 and −9, its polynomial is x^2-9. (1 mark)

Answer: True.

Explanation: For a monic quadratic, $p(x)=x^2-(\text{sum})x+(\text{product})$. Substituting sum=0 and product=−9 gives x^2-9.

Q5. [Word problem · Moderate] Two consecutive positive integers have product 72. Form a quadratic equation and find the integers. (1 mark)

Answer: The integers are 8 and 9.

Explanation: Let the smaller integer be x. Then $x(x+1)=72$, so $x^2+x-72=0$. Factorising gives $(x-8)(x+9)=0$. Since the integers are positive, x=8. The integers are 8 and 9.

Q6. [PYQ-style · Moderate] If α,β are zeros of x^2-6x+4, find $1/\alpha+1/\beta$. (1 mark)

Answer: 3/2.

Explanation: Use $1/\alpha+1/\beta=(\alpha+\beta)/(\alpha\beta)$. Here $\alpha+\beta=6$ and $\alpha\beta=4$. Hence the value is 6/4=3/2.

Q7. [Choose all that apply · Moderate] For x^2-4x+4, which statements are true? (A) It has two distinct zeros (B) It has a repeated zero (C) Its zero is 2 (D) Sum of zeros is 4 (1 mark)

Correct answers: (B), (C) and (D).

Explanation: $x^2-4x+4=(x-2)^2$, so the repeated zero is 2. Counting multiplicity, the zeros are 2 and 2, and their sum is 4. Hence B, C and D are true.

Q8. [Challenge · Hard] If α,β are zeros of $3x^2-5x-2$, find $(\alpha-\beta)^2$. (3 marks)

Answer: 49/9.

Explanation: Here $\alpha+\beta=5/3$ and $\alpha\beta=-2/3$. Use $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$. Therefore $(5/3)^2-4(-2/3)=25/9+24/9=49/9$.

Q9. [Subjective · 3 marks] Find a quadratic polynomial whose zeros are $2+\sqrt{3}$ and $2-\sqrt{3}$. (3 marks)

Answer: x^2-4x+1.

Explanation: Sum of zeros=$(2+\sqrt{3})+(2-\sqrt{3})=4$. Product=$(2+\sqrt{3})(2-\sqrt{3})=4-3=1$. Hence the monic polynomial is x^2-4x+1.

Q10. [Subjective · 4 marks] A rectangle has area $x^2-2x-15$. Factorise the polynomial and identify its zeros. Explain why the negative zero is not a possible side length if x represents a positive length. (4 marks)

Answer: Factorisation = $(x-5)(x+3)$; zeros are 5 and −3.

Explanation: Since $x^2-2x-15=(x-5)(x+3)$, setting each factor to zero gives x=5 and x=−3. If x represents a physical length, it must be positive. Therefore −3 is not a feasible length, even though it is an algebraic zero.

MOCK SET 03 | Parameter practice

Q1. [Rapid fire · Easy] State the degree of $4x^{\blacksquare}-2x^2+x$. (1 mark)

Answer: 7.

Explanation: The highest exponent with a non-zero coefficient is 7.

Q2. [MCQ · Easy] A zero of $7x+21$ is: (A) 3 (B) -3 (C) 7 (D) -7 (1 mark)

Correct answer: (B) -3

Explanation: Set $7x+21=0$. Then $7x=-21$, so $x=-3$.

Q3. [Coefficient link · Moderate] For $5x^2+px+10$, the product of zeros is 2. What is p? (1 mark)

Answer: p cannot be determined from the given information.

Explanation: For ax^2+bx+c, product of zeros=c/a. Here it is $10/5=2$ for every value of p. So the condition gives no information about p.

Q4. [Correct the statement · Moderate] A student writes: for ax^2+bx+c, sum of zeros is b/a. Correct it. (1 mark)

Answer: The sum of zeros is $-b/a$.

Explanation: For a quadratic ax^2+bx+c, the relationship between the zeros α and β is $\alpha+\beta=-b/a$. The negative sign is essential.

Q5. [Word problem · Moderate] A square courtyard's area expression is $x^2-12x+36$. Find the zero and explain its multiplicity. (1 mark)

Answer: 6, a repeated zero.

Explanation: $x^2-12x+36=(x-6)^2$. Thus $x=6$ is the only distinct zero and it occurs twice, so it is a repeated zero.

Q6. [PYQ-theme · Moderate] If one zero of $2x^2-5x+k$ is 2, find k and the other zero. (1 mark)

Answer: k=2; the other zero is 1/2.

Explanation: Substitute $x=2$: $2(2^2)-5(2)+k=0$, so $8-10+k=0$ and $k=2$. The product of zeros is $k/2=1$. Since one zero is 2, the other is 1/2.

Q7. [Assertion–Reason · Hard] Assertion: A quadratic polynomial can have at most two real zeros. Reason: A non-zero polynomial of degree n has at most n zeros. Choose: (A) both true, reason explains (B) both true, no explanation (C) A true R false (D) A false R true. (1 mark)

Correct answer: (A) Both are true, and the reason explains the assertion.

Explanation: A quadratic has degree 2. The general result says a non-zero polynomial of degree n has at most n zeros. Taking n=2 proves that a quadratic has at most two real zeros.

Q8. [Challenge · Hard] If α,β are zeros of x^2+3x-7, evaluate $\alpha^2+\beta^2+\alpha\beta$. (3 marks)

Answer: 16.

Explanation: Here $\alpha+\beta=-3$ and $\alpha\beta=-7$. First, $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=9+14=23$. Therefore $\alpha^2+\beta^2+\alpha\beta=23-7=16$.

Q9. [Subjective · 3 marks] Find zeros of $6x^2+x-2$ and verify their product. (3 marks)

Answer: 1/2 and $-2/3$.

Explanation: Factorise: $6x^2+x-2=(3x+2)(2x-1)$. Thus the zeros are $-2/3$ and $1/2$. Their product is $(-2/3)(1/2)=-1/3$, which equals $c/a=-2/6=-1/3$.

Q10. [Subjective · 4 marks] A quadratic polynomial has zeros α,β with $\alpha+\beta=-4$ and $\alpha\beta=-5$. Form the polynomial, then find its zeros. (4 marks)

Answer: x^2+4x-5; zeros are 1 and -5.

Explanation: The monic polynomial is $x^2-(\alpha+\beta)x+\alpha\beta=x^2-(-4)x-5=x^2+4x-5$. Factorising gives $(x+5)(x-1)$, so the zeros are -5 and 1.

MOCK SET 04 | Reasoning clinic

Q1. [Visual interpretation · Easy] A parabola meets the x-axis at x=–3 and x=1. What are its zeros? (1 mark)

Answer: –3 and 1.

Explanation: The x-intercepts of the graph give the zeros of the polynomial.

Q2. [MCQ · Moderate] For $x^2+px+16$ with equal zeros, p can be: (A) 8 only (B) –8 only (C) ±8 (D) ±16 (1 mark)

Correct answer: (C) ±8

Explanation: Equal zeros require the discriminant to be zero. $p^2–4(1)(16)=0$, so $p^2=64$ and $p=±8$.

Q3. [Form polynomial · Moderate] Write a quadratic polynomial with zeros 0 and –6. (1 mark)

Answer: x^2+6x.

Explanation: Sum=0+(–6)=–6 and product=0. The monic polynomial is $x^2–(–6)x+0=x^2+6x$.

Q4. [Think carefully · Moderate] Can a quadratic polynomial have exactly one distinct real zero? Give an example. (1 mark)

Answer: Yes. Example: $x^2–6x+9$.

Explanation: $x^2–6x+9=(x–3)^2$. It has one distinct real zero, 3, repeated twice.

Q5. [Word problem · Moderate] The product of two consecutive integers is 156. Find the integers by forming a polynomial. (1 mark)

Answer: 12 and 13 (also –13 and –12).

Explanation: Let the smaller integer be x. Then $x(x+1)=156$, so $x^2+x–156=0=(x–12)(x+13)$. Thus x=12 or x=–13. The corresponding pairs are 12,13 and –13,–12.

Q6. [PYQ-style · Moderate] If α,β are zeros of $2x^2–3x–5$, evaluate $\alpha^2\beta+\alpha\beta^2$. (1 mark)

Answer: 15/4.

Explanation: Factorise or use the relations: $\alpha+\beta=3/2$ and $\alpha\beta=–5/2$. Then $\alpha^2\beta+\alpha\beta^2=\alpha\beta(\alpha+\beta)=(–5/2)(3/2)=–15/4$.

Q7. [Error analysis · Hard] A learner claims zeros 2 and 5 give $x^2+7x+10$. Identify the mistake and write the correct polynomial. (1 mark)

Answer: The correct polynomial is $x^2–7x+10$.

Explanation: The sum of the zeros is 2+5=7 and the product is 10. The monic polynomial is $x^2–(\text{sum})x+(\text{product})=x^2–7x+10$. The learner used the wrong sign for the middle term.

Q8. [Challenge · Hard] If α,β are zeros of $4x^2–4x–3$, find $\alpha/\beta+\beta/\alpha$. (3 marks)

Answer: –10/3.

Explanation: Here $\alpha+\beta=1$ and $\alpha\beta=–3/4$. Use $\alpha/\beta+\beta/\alpha=(\alpha^2+\beta^2)/(\alpha\beta)$. Also $\alpha^2+\beta^2=(\alpha+\beta)^2–2\alpha\beta=1–2(–3/4)=5/2$. Hence $(5/2)/(–3/4)=–10/3$.

Q9. [Subjective · 3 marks] Find the zeros of $x^2–2\sqrt{2}x+2$ and state their nature. (3 marks)

Answer: $\sqrt{2}$, repeated.

Explanation: $x^2–2\sqrt{2}x+2=(x–\sqrt{2})^2$. Hence the polynomial has one distinct real zero $\sqrt{2}$ with multiplicity 2.

Q10. [Subjective · 4 marks] A quadratic polynomial has one zero 4 and the sum of its zeros is –1. Find its other zero and a monic polynomial. (4 marks)

Answer: The other zero is –5; polynomial is $x^2+x–20$.

Explanation: Let the other zero be β. Then $4+\beta=–1$, so $\beta=–5$. The product is $4(–5)=–20$. The monic polynomial is $x^2–(–1)x–20=x^2+x–20$.

MOCK SET 05 | Relations toolbox

Q1. [MCQ · Easy] Which is a quadratic polynomial? (A) x^3+1 (B) $5x^2-2x+1$ (C) $4x+3$ (D) 7 (1 mark)

Correct answer: (B) $5x^2-2x+1$

Explanation: A quadratic polynomial has degree 2. Option B has highest exponent 2.

Q2. [Short response · Easy] For $p(x)=x^2-1$, calculate $p(-1)$. Is -1 a zero? (1 mark)

Answer: $p(-1)=0$; yes, -1 is a zero.

Explanation: Substitute $x=-1$: $p(-1)=(-1)^2-1=1-1=0$. A value is a zero when it makes the polynomial equal to zero.

Q3. [Use relations · Moderate] For $2x^2+9x+4$, find sum and product of zeros. (1 mark)

Answer: Sum = $-9/2$; product = 2.

Explanation: Here $a=2$, $b=9$ and $c=4$. Sum$=-b/a=-9/2$; product$=c/a=4/2=2$.

Q4. [True/False · Moderate] If $\alpha+\beta=5$ and $\alpha\beta=6$, then x^2-5x+6 has zeros α and β. (1 mark)

Answer: True.

Explanation: The monic polynomial formed from zeros α and β is $x^2-(\alpha+\beta)x+\alpha\beta$. Substituting 5 and 6 gives x^2-5x+6.

Q5. [Word problem · Moderate] A rectangular banner's area polynomial is $x^2-9x+20$. Its dimensions are $x-4$ and $x-5$. Find its zeros and interpret them as boundary values. (1 mark)

Answer: 4 and 5.

Explanation: $x^2-9x+20=(x-4)(x-5)$. Setting each factor to zero gives $x=4$ and $x=5$. These are the boundary values where one of the dimension expressions becomes zero.

Q6. [PYQ-theme · Moderate] If α,β are zeros of $x^2-7x+10$, find $\alpha^2+\beta^2$. (1 mark)

Answer: 29.

Explanation: Sum=7 and product=10. Thus $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=7^2-2(10)=49-20=29$.

Q7. [Assertion–Reason · Hard] Assertion: If the product of zeros of ax^2+bx+c is zero, at least one zero is zero. Reason: The product of zeros equals c/a. (A) both true and reason explains (B) both true but no explanation (C) A true R false (D) A false R true (1 mark)

Correct answer: (A) Both are true, and the reason explains the assertion.

Explanation: For ax^2+bx+c, the product of zeros is c/a. If this product is zero, then $\alpha\beta=0$, which means at least one of α or β is zero.

Q8. [Challenge · Hard] If α,β are zeros of x^2-5x+1, find $(\alpha-1)(\beta-1)$. (3 marks)

Answer: -3.

Explanation: Here $\alpha+\beta=5$ and $\alpha\beta=1$. Expand: $(\alpha-1)(\beta-1)=\alpha\beta-\alpha-\beta+1=1-5+1=-3$.

Q9. [Subjective · 3 marks] Find zeros of $3x^2-11x-4$ and verify sum and product. (3 marks)

Answer: 4 and $-1/3$.

Explanation: Factorise: $3x^2-11x-4=(3x+1)(x-4)$. Thus zeros are $-1/3$ and 4. Sum$=4-1/3=11/3=-b/a$. Product$=4(-1/3)=-4/3=c/a$.

Q10. [Subjective · 4 marks] Find a quadratic polynomial whose zeros are reciprocals of the zeros of $2x^2-5x+3$. (4 marks)

Answer: $3x^2-5x+2$.

Explanation: Factorise $2x^2-5x+3=(2x-3)(x-1)$, so its zeros are 3/2 and 1. Their reciprocals are 2/3 and 1. A polynomial with these zeros is $(3x-2)(x-1)=3x^2-5x+2$.

MOCK SET 06 | Application & logic

Q1. [Concept check · Easy] What is the zero of a polynomial p(x) called? (1 mark)

Answer: A value r for which p(r)=0.

Explanation: By definition, r is a zero of p(x) if substituting x=r makes p(r)=0.

Q2. [MCQ · Easy] The zeros of x^2+2x-8 are: (A) 2,–4 (B) –2,4 (C) 1,–8 (D) 2,4 (1 mark)

Correct answer: (A) 2, –4

Explanation: Factorise: $x^2+2x-8=(x+4)(x-2)$. Therefore the zeros are –4 and 2.

Q3. [Parameter · Moderate] If 3 is a zero of $x^2-(k+1)x+12$, find k. (1 mark)

Answer: k=6.

Explanation: Substitute x=3: $9-3(k+1)+12=0$. Thus $21-3k-3=0$, so $18-3k=0$ and k=6.

Q4. [Short proof · Moderate] Explain why a non-zero linear polynomial has exactly one real zero. (1 mark)

Answer: A non-zero linear polynomial ax+b, where $a\neq0$, has exactly one real zero.

Explanation: Solving ax+b=0 gives x=–b/a. Since a is non-zero, this gives one unique real value.

Q5. [Word problem · Moderate] Two consecutive odd positive integers have product 143. Find them using a quadratic polynomial. (1 mark)

Answer: 11 and 13.

Explanation: Let the smaller odd integer be x. Then the next odd integer is x+2. So $x(x+2)=143$, giving $x^2+2x-143=0=(x-11)(x+13)$. Since the integers are positive, x=11. The integers are 11 and 13.

Q6. [PYQ-style · Moderate] For zeros α,β of $3x^2-2x-1$, evaluate $\alpha^2+\beta^2$. (1 mark)

Answer: 10/9.

Explanation: Sum=2/3 and product=–1/3. Therefore $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=4/9+2/3=4/9+6/9=10/9$.

Q7. [Choose the valid conclusion · Hard] For $p(x)=x^2+4x+8$, the graph: (A) cuts x-axis twice (B) touches once (C) has no x-intercept (D) crosses at x=0 (1 mark)

Correct answer: (C) has no x-intercept

Explanation: Complete the square: $x^2+4x+8=(x+2)^2+4$. This is always positive for real x, so the graph never meets the x-axis.

Q8. [Challenge · Hard] If α,β are zeros of x^2-4x-1, evaluate $\alpha^3+\beta^3$. (3 marks)

Answer: 76.

Explanation: Here $\alpha+\beta=4$ and $\alpha\beta=-1$. Use $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)=4^3-3(-1)(4)=64+12=76$.

Q9. [Subjective · 3 marks] Find the zeros of $2x^2+5x-3$ and verify the sum relation. (3 marks)

Answer: 1/2 and –3.

Explanation: Factorise: $2x^2+5x-3=(2x-1)(x+3)$. Thus the zeros are 1/2 and –3. Their sum is $1/2-3=-5/2$, which equals $-b/a=-5/2$.

Q10. [Subjective · 4 marks] If α,β are zeros of $x^2-px+12$ and $\alpha^2+\beta^2=25$, find possible p values. (4 marks)

Answer: p=±7.

Explanation: Since $\alpha+\beta=p$ and $\alpha\beta=12$, $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=p^2-24$. Given $p^2-24=25$, so $p^2=49$. Therefore p=7 or p=–7.

MOCK SET 07 | Algebraic thinking

Q1. [MCQ · Easy] The graph of $p(x)=x^2+1$ has how many x-intercepts? (A) 0 (B) 1 (C) 2 (D) 3 (1 mark)

Correct answer: (A) 0

Explanation: For every real x, $x^2 \geq 0$, so x^2+1 is always positive. The graph has no x-intercepts.

Q2. [Fast factorisation · Easy] Factorise $x^2-13x+40$. (1 mark)

Answer: $(x-5)(x-8)$.

Explanation: Two numbers with product 40 and sum −13 are −5 and −8. Hence $x^2-13x+40=(x-5)(x-8)$.

Q3. [Relation · Moderate] For $4x^2+4x-3$, calculate sum and product of zeros. (1 mark)

Answer: Sum = −1; product = −3/4.

Explanation: Here a=4, b=4 and c=−3. Sum=−b/a=−4/4=−1; product=c/a=−3/4.

Q4. [Fix the solution · Moderate] A student says x^2-16 has only one zero, 4. Correct the answer. (1 mark)

Answer: The zeros are 4 and −4.

Explanation: $x^2-16=(x-4)(x+4)$. Setting both factors to zero gives x=4 and x=−4.

Q5. [Word problem · Moderate] A square picture's area is modelled by x^2-16. Factorise and state its zeros. Which root may represent a positive side length? (1 mark)

Answer: Factorisation = $(x-4)(x+4)$; zeros are 4 and −4. The positive side length is 4.

Explanation: Using the difference of two squares, $x^2-16=(x-4)(x+4)$. The zeros are 4 and −4. A physical side length must be positive, so 4 is the feasible value.

Q6. [PYQ-theme · Moderate] If α,β are zeros of $5x^2-7x+2$, find $1/\alpha+1/\beta$. (1 mark)

Answer: 7/2.

Explanation: Sum=7/5 and product=2/5. Thus $1/\alpha+1/\beta=(\alpha+\beta)/(\alpha\beta)=(7/5)/(2/5)=7/2$.

Q7. [Assertion–Reason · Hard] Assertion: A quadratic can have a repeated zero. Reason: Its graph may touch the x-axis at its vertex. (A) both true, reason explains (B) both true, no explanation (C) A true R false (D) A false R true (1 mark)

Correct answer: (A) Both are true, and the reason explains the assertion.

Explanation: A quadratic can have equal roots when its discriminant is zero. In that case, its parabola touches the x-axis at its vertex rather than crossing it.

Q8. [Challenge · Hard] If α,β are zeros of $2x^2+x-4$, find $(\alpha+1)(\beta+1)$. (3 marks)

Answer: −3/2.

Explanation: Here $\alpha+\beta=-1/2$ and $\alpha\beta=-2$. Expand: $(\alpha+1)(\beta+1)=\alpha\beta+\alpha+\beta+1=-2-1/2+1=-3/2$.

Q9. [Subjective · 3 marks] Find the zeros of $5x^2-13x+6$ and verify the product relation. (3 marks)

Answer: 2 and 3/5.

Explanation: Factorise: $5x^2-13x+6=(5x-3)(x-2)$. Thus the zeros are 3/5 and 2. Their product is $(3/5)(2)=6/5=c/a=6/5$.

Q10. [Subjective · 4 marks] A quadratic polynomial has zeros whose sum is 2 and product is −8. Form the polynomial, solve it, and verify both relations. (4 marks)

Answer: x^2-2x-8; zeros are 4 and −2.

Explanation: The monic polynomial is x^2-2x-8. Factorise: $(x-4)(x+2)=0$, so the zeros are 4 and −2. Their sum is $4+(-2)=2$ and product is $4(-2)=-8$, as required.

MOCK SET 08 | Error detection

Q1. [MCQ · Easy] If p(–2)=0, then –2 is: (A) a coefficient (B) a zero (C) the degree (D) the constant term (1 mark)

Correct answer: (B) a zero

Explanation: A number r is a zero of p(x) when p(r)=0. Since p(–2)=0, –2 is a zero.

Q2. [Mixed skill · Easy] Find the zeros of x^2–25. (1 mark)

Answer: –5 and 5.

Explanation: x^2–25=(x–5)(x+5). Setting each factor to zero gives x=5 and x=–5.

Q3. [Relation · Moderate] If α,β are zeros of $2x^2$–3x–9, find $\alpha\beta$ and $\alpha+\beta$. (1 mark)

Answer: $\alpha\beta$=–9/2; $\alpha+\beta$=3/2.

Explanation: Here a=2, b=–3 and c=–9. Product=c/a=–9/2 and sum=–b/a=3/2.

Q4. [Short response · Moderate] Can two different quadratic polynomials have the same zeros? Explain briefly. (1 mark)

Answer: Yes.

Explanation: One polynomial can be a non-zero constant multiple of another. For example, x^2–5x+6 and $2x^2$–10x+12 have the same zeros, 2 and 3.

Q5. [Word problem · Moderate] A rectangular plot has area x^2–x–12, with side expressions x–4 and x+3. Find its zeros and state the positive boundary root. (1 mark)

Answer: 4 and –3; positive boundary root = 4.

Explanation: x^2–x–12=(x–4)(x+3). Setting each factor to zero gives x=4 and x=–3. The positive boundary value is 4.

Q6. [PYQ-style · Moderate] If α,β are zeros of x^2+2x–3, find $\alpha^2\beta+\alpha\beta^2$. (1 mark)

Answer: 6.

Explanation: Here $\alpha+\beta$=–2 and $\alpha\beta$=–3. Then $\alpha^2\beta+\alpha\beta^2=\alpha\beta(\alpha+\beta)$=(–3)(–2)=6.

Q7. [Reasoning · Hard] A monic quadratic has integer zeros, sum 1 and product –12. Find the zeros and explain how the signs are determined. (1 mark)

Answer: 4 and –3.

Explanation: The product is negative, so the zeros have opposite signs. The integer pair with product –12 and sum 1 is 4 and –3.

Q8. [Challenge · Hard] If α,β are zeros of x^2–2x–1, find $\alpha^4+\beta^4$. (3 marks)

Answer: 34.

Explanation: Here $\alpha+\beta$=2 and $\alpha\beta$=–1. First $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$=4+2=6. Then $\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2(\alpha\beta)^2=6^2-2(1)$=34.

Q9. [Subjective · 3 marks] Find zeros of $2x^2$–9x+10 and verify coefficient relations. (3 marks)

Answer: 2 and 5/2.

Explanation: Factorise: $2x^2$–9x+10=(2x–5)(x–2). Thus zeros are 5/2 and 2. Sum=9/2=–b/a and product=5=c/a.

Q10. [Subjective · 4 marks] A quadratic polynomial has zeros α,β. If $\alpha+\beta$=3 and $\alpha^2+\beta^2$=13, form the monic polynomial and find its zeros. (4 marks)

Answer: x^2–3x–2; zeros are (3+$\sqrt{17}$)/2 and (3–$\sqrt{17}$)/2.

Explanation: Use $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$. Thus 13=9–2$\alpha\beta$, so $\alpha\beta$=–2. The monic polynomial is x^2–3x–2. Solving x^2–3x–2=0 using the quadratic formula gives x=(3±$\sqrt{17}$)/2.

MOCK SET 09 | Higher-order mix

Q1. [MCQ · Easy] Which polynomial has a repeated zero? (A) x^2-5x+6 (B) $x^2-10x+25$ (C) x^2-1 (D) x^2+3x+2 (1 mark)

Correct answer: (B) $x^2-10x+25$

Explanation: $x^2-10x+25=(x-5)^2$, so it has the repeated zero 5.

Q2. [Graph interpretation · Easy] If a parabola lies completely above the x-axis, what can you say about its real zeros? (1 mark)

Answer: It has no real zeros.

Explanation: Real zeros are the x-values where the graph intersects the x-axis. A graph completely above the x-axis never intersects it.

Q3. [Build it · Moderate] Find a quadratic polynomial with zeros -4 and $1/2$ and integer coefficients, choosing the simplest non-zero multiple. (1 mark)

Answer: $2x^2+7x-4$.

Explanation: Using the zeros, form $(x+4)(x-1/2)=x^2+(7/2)x-2$. Multiplying by 2 to obtain integer coefficients gives $2x^2+7x-4$.

Q4. [Reasoning · Moderate] For x^2-6x+q, one zero is twice the other. Find q. (1 mark)

Answer: q=8.

Explanation: Let the zeros be t and 2t. Their sum is 6, so 3t=6 and t=2. The product is $t(2t)=4\times$? Specifically $2\times4=8$, hence q=8.

Q5. [Word problem · Moderate] A rectangular mural has area $x^2-3x-10$. Its side factors are $x-5$ and $x+2$. Find zeros and discuss which values can be valid if x is a positive dimension parameter. (1 mark)

Answer: Zeros are 5 and -2; valid dimensions require $x>5$.

Explanation: $x^2-3x-10=(x-5)(x+2)$, so the zeros are 5 and -2. For both dimensions $x-5$ and $x+2$ to be positive, x must be greater than 5. The roots mark boundary values, not feasible interior dimensions.

Q6. [PYQ-theme · Moderate] If α,β are zeros of $2x^2-x-6$, evaluate $(1/\alpha)+(1/\beta)$. (1 mark)

Answer: $-1/6$.

Explanation: Sum=1/2 and product=-3. Thus $1/\alpha+1/\beta=(\alpha+\beta)/(\alpha\beta)=(1/2)/(-3)=-1/6$.

Q7. [Higher-order · Hard] If α,β are zeros of $3x^2-4x-2$, find $\alpha^3+\beta^3$. (1 mark)

Answer: 136/27.

Explanation: Here $\alpha+\beta=4/3$ and $\alpha\beta=-2/3$. Use $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)=(4/3)^3-3(-2/3)(4/3)=64/27+8/3=64/27+72/27=136/27$.

Q8. [Challenge · Hard] A monic quadratic has zeros α,β, with $\alpha+\beta=-2$ and $\alpha\beta=-8$. Evaluate $\alpha^2\beta^2-(\alpha+\beta)^2$. (3 marks)

Answer: 60.

Explanation: Since $\alpha\beta=-8$, $\alpha^2\beta^2=(\alpha\beta)^2=64$. Also $(\alpha+\beta)^2=(-2)^2=4$. Therefore the required value is $64-4=60$.

Q9. [Subjective · 3 marks] Find zeros of $x^2-7x+12$ and verify both coefficient relationships. (3 marks)

Answer: 3 and 4.

Explanation: Factorise: $x^2-7x+12=(x-3)(x-4)$. The zeros are 3 and 4. Their sum is $7=-b/a$ and their product is $12=c/a$.

Q10. [Subjective · 5 marks] The zeros of a quadratic polynomial are α,β, where $\alpha+\beta=-3$ and $\alpha\beta=-10$. (i) Form its monic polynomial. (ii) Find the zeros. (iii) Verify the relations. (4 marks)

Answer: $x^2+3x-10$; zeros are 2 and -5.

Explanation: The monic polynomial is $x^2-(-3)x+(-10)=x^2+3x-10$. Factorising gives $(x+5)(x-2)=0$, so the zeros are -5 and 2. Their sum is -3 and product is -10, verifying both relations.

MOCK SET 10 | Final challenge

Q1. [Warm-up · Easy] For $p(x)=2x^2-3x+1$, calculate $p(1)$. (1 mark)

Answer: $p(1)=0$.

Explanation: Substitute $x=1$: $p(1)=2(1)^2-3(1)+1=2-3+1=0$.

Q2. [MCQ · Easy] The zeros of x^2+x-6 are: (A) 2,−3 (B) −2,3 (C) 1,−6 (D) 2,3 (1 mark)

Correct answer: (A) 2, −3

Explanation: Factorise: $x^2+x-6=(x+3)(x-2)$. Thus the zeros are -3 and 2.

Q3. [Connect concepts · Moderate] A monic quadratic has zeros −1 and 6. Find its sum, product and polynomial. (1 mark)

Answer: Sum = 5; product = −6; polynomial = x^2-5x-6.

Explanation: Sum$=-1+6=5$ and product$=(-1)(6)=-6$. The monic polynomial is x^2-5x-6.

Q4. [Diagnose · Moderate] A learner finds the product of zeros of $7x^2-2x-3$ as −3. Correct the answer. (1 mark)

Answer: −3/7.

Explanation: For ax^2+bx+c, product of zeros$=c/a$. Here $c=-3$ and $a=7$, so the product is $-3/7$, not -3.

Q5. [Word problem · Moderate] A community garden's area expression is $x^2+3x-28$ and factors as $(x+7)(x-4)$. Find the zeros. Which value is a positive boundary? (1 mark)

Answer: −7 and 4; positive boundary = 4.

Explanation: Set $x+7=0$ and $x-4=0$. The zeros are -7 and 4. The positive boundary value is 4.

Q6. [PYQ-style · Moderate] If α,β are zeros of $4x^2-4x-3$, find $(\alpha+\beta)^2-2\alpha\beta$. (1 mark)

Answer: 5/2.

Explanation: Here $\alpha+\beta=1$ and $\alpha\beta=-3/4$. Thus $(\alpha+\beta)^2-2\alpha\beta=1-2(-3/4)=1+3/2=5/2$.

Q7. [Higher-order · Hard] If α,β are zeros of x^2-3x+1, find $\alpha^3+\beta^3$. (1 mark)

Answer: 18.

Explanation: Sum$=3$ and product$=1$. Therefore $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)=27-9=18$.

Q8. [Challenge · Hard] If α,β are zeros of $2x^2+5x-3$, find $\alpha^2\beta^2+\alpha\beta$. (3 marks)

Answer: 3/4.

Explanation: Here $\alpha\beta=c/a=-3/2$. Thus $\alpha^2\beta^2+\alpha\beta=(\alpha\beta)^2+\alpha\beta=(-3/2)^2-3/2=9/4-6/4=3/4$.

Q9. [Subjective · 3 marks] Find the zeros of $4x^2+4x-3$ and verify the sum and product relations. (3 marks)

Answer: 1/2 and −3/2.

Explanation: Factorise: $4x^2+4x-3=(2x-1)(2x+3)$. Thus zeros are $1/2$ and $-3/2$. Their sum is $-1=-b/a=-4/4$, and their product is $-3/4=c/a$.

Q10. [Subjective · 5 marks] A quadratic polynomial has zeros α and β such that $\alpha+\beta=5$ and $\alpha^2+\beta^2=13$. Find (i) $\alpha\beta$, (ii) the monic polynomial, and (iii) its zeros. (5 marks)

Answer: $\alpha\beta=6$; polynomial = x^2-5x+6; zeros are 2 and 3.

Explanation: Use $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$. Thus $13=25-2\alpha\beta$, giving $2\alpha\beta=12$ and $\alpha\beta=6$. The monic polynomial is $x^2-5x+6=(x-2)(x-3)$. Therefore its zeros are 2 and 3.